SD Street Safety

ECE 143 Group 3

Dec 3, 2025

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Motivation

- Improve public safety by identifying infrastructure in most need of repair
- Encourage caution in high-risk traffic areas
- Inform city officials and the public of problem areas
- Raise awareness of the dangers of aging infrastructure

Datasets (from city portal)

- Traffic collisions - basic reports
- Zoning
- Traffic Volumes
- Streets Repair Line Segments
- Pavement Condition Assessments
- Streets Repair Projects

Streets Repair Line Segments

Shapefile of the City's street network. This shapefile contains the street segments identified in our streets repair projects and Overall Condition Index datasets. This data is displayed on streets.sandiego.gov.

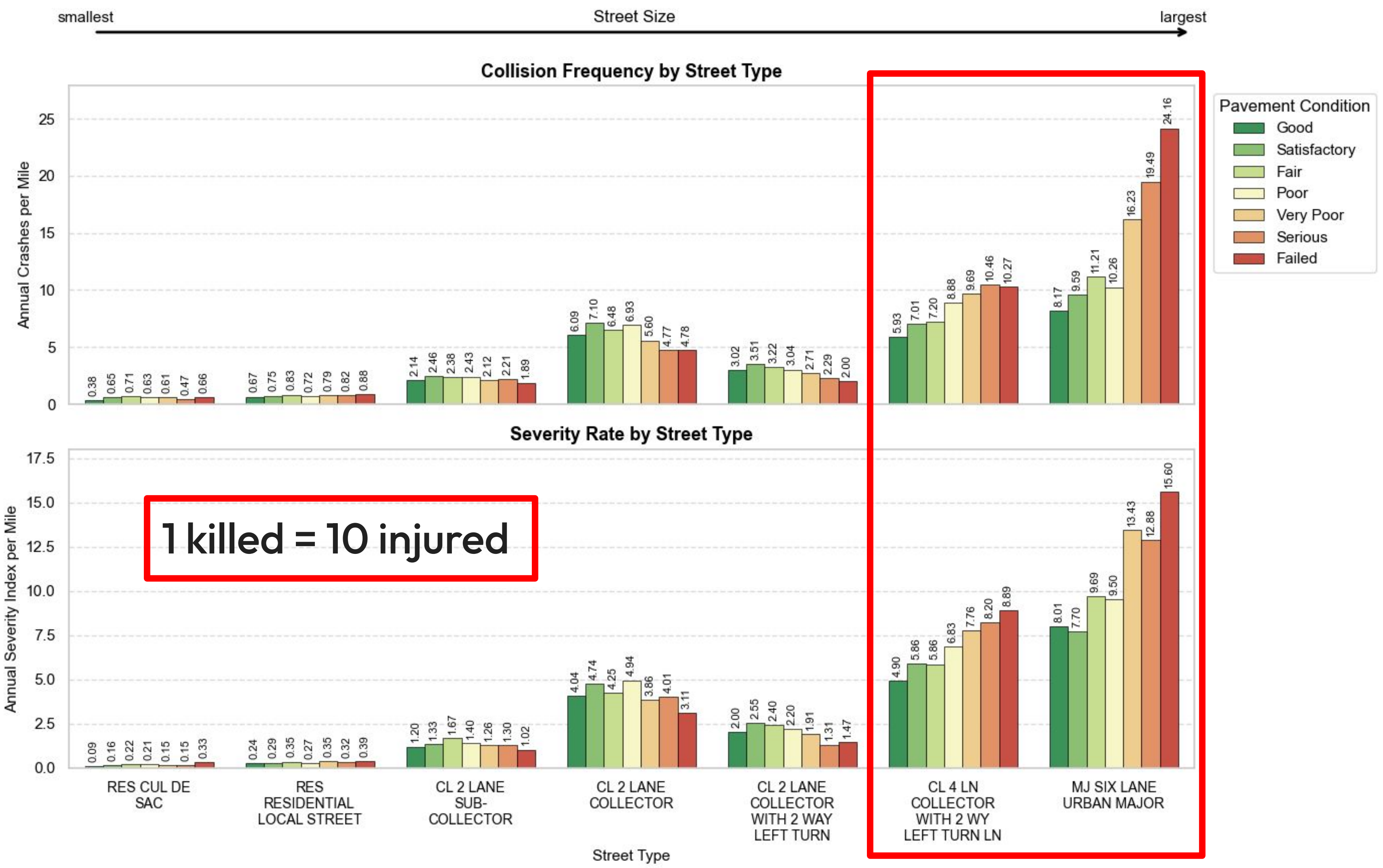
Resources

RESOURCE	TYPE	DOWNLOAD	PREVIEW
Street segment lines	shp		
Street segment lines	topojson		
Street segment lines	geojson		
Street segment attribute table	csv		

OBJECTID	ROADSEGID	IAMFLOC	RD20FULL	XSTRT1	XSTRT2	LLOWADDR	LHIGHADDR	RLOWADDR	RHIGHADDR	ZIP
26001	931757	SS-027187-PV1	UNIVERSITY AV	VAN DYKE AV	43RD ST	4250	4298	4251	4299	92105
20118	919561	SS-029653-PV1	S SAN JACINTO DR	MANZANARES WY	SAN JACINTO PL	201	211	200	210	92114
15042	917267	SS-023539-PV1	ROSECRANS ST	MADRID ST	NORTH EVERGREEN ST	3200	3248	3201	3249	92110
13081	927537	SS-016508-PV1	LINDA VISTA RD	NAPA ST	METRO ST	5200	5258	5201	5259	92110
9159	907894	SS-017300-PV1	MALTA ST	ATOLL ST	BELFORD ST	7100	7198	7101	7199	92111
7198	906334	SS-019442-PV1	MT ST HELENS DR	BEGIN	MT CRESTI DR	4700	4748	4701	4749	92117
18862	930137	SS-000181-PV1	05TH AV	F ST	E ST	800	898	801	899	92101
12376	926929	SS-011801-PV1	GALVESTON ST	FRANKFORT ST	LITTLEFIELD ST	1400	1598	1401	1599	92110
10415	926056	SS-012723-PV1	GREYLING DR	GERALDINE AV	PINECREST AV	2920	2948	2921	2949	92123
8454	907292	SS-015547-PV1	LA JOLLA HERMOSA AV	BIRD ROCK AV	ALTA WY	5700	5798	5701	5799	92037
15593	917463	SS-016653-PV1	LOCUST ST	IBSEN ST	JAMES ST	2850	2898	2851	2899	92106
9710	908334	SS-028778-PV1	WILBUR AV	MISSION BL	BAYARD ST	800	898	801	899	92109
811	916008	SS-016704-PV1	LOFTY TL CT	BEGIN	LOFTY TL DR	15500	15548	15501	15549	92127
24693	913306	SS-007142-PV1	CHICO ST	SHASTA ST	KENDALL ST	1700	1798	1701	1799	92109
18810	918654	SS-000848-PV1	34TH ST	E ST	PICKWICK ST	850	948	851	949	92102

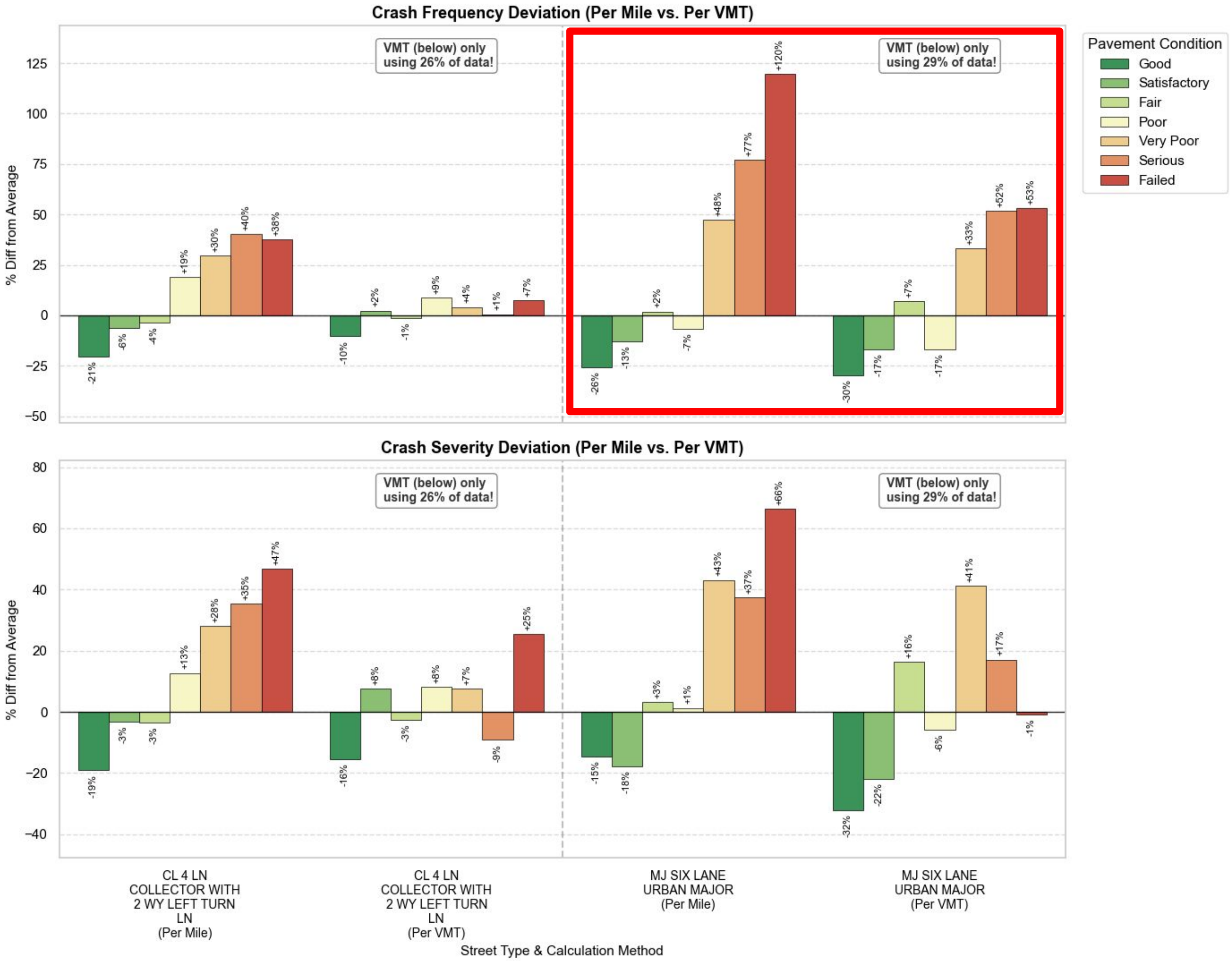
How does pavement condition impact street safety?

- I **analyzed each street type separately** due to their distinct characteristics
- Analysis required linking **pavement condition**, **collisions**, **repair segments**, & **traffic volumes** data
- Traffic volume data is **limited**!

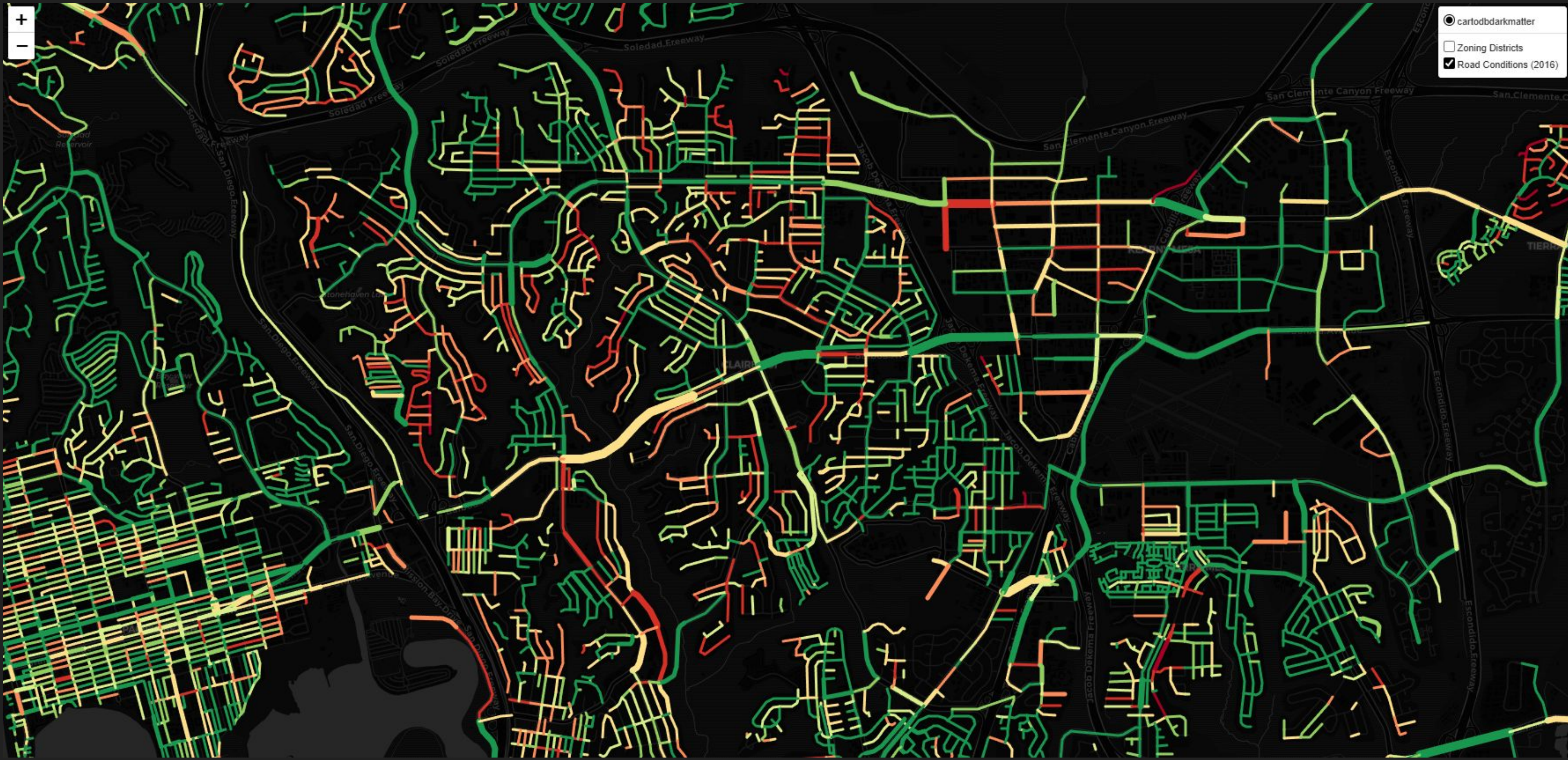


smallest
Street Size
largest
Collision Frequency by Street Type
Annual Crashes per Mile
Severity Rate by Street Type
Annual Severity Index per Mile
Pavement Condition
Good
Satisfactory
Fair
Poor
Very Poor
Serious
Failed
1 killed = 10 injured
RES CUL DE SAC
RES RESIDENTIAL LOCAL STREET
CL 2 LANE SUB-COLLECTOR
CL 2 LANE COLLECTOR
CL 2 LANE COLLECTOR WITH 2 WAY LEFT TURN
CL 4 LN COLLECTOR WITH 2 WY LEFT TURN LN
MJ SIX LANE URBAN MAJOR
Street Type


Only using ~17.6% of original data! (lacking sufficient traffic data)
smallest
Street Size
largest
Collision Frequency by Street Type
Crashes per Million VMT
Severity Rate by Street Type
Severity Index per Million VMT
Pavement Condition
Good
Satisfactory
Fair
Poor
Very Poor
Serious
Failed
CL 2 LANE SUB-COLLECTOR
CL 2 LANE COLLECTOR
CL 2 LANE COLLECTOR WITH 2 WAY LEFT TURN
CL 4 LN COLLECTOR WITH 2 WY LEFT TURN LN
MJ SIX LANE URBAN MAJOR
Street Type
2.57 2.43 2.24 1.86 2.27 2.55 1.58
2.63 3.26 2.73 2.52 1.60 1.58 1.97
1.31 1.73 2.19 1.00 0.98 1.27 1.40
1.40 1.60 1.54 1.69 1.62 1.57 1.67
0.65 0.77 1.00 0.77 1.24 1.41 1.42
1.55 1.39 1.55 1.29 1.12 1.44 0.67
1.63 2.31 1.89 1.91 1.11 1.57 1.24
0.82 1.23 1.88 0.76 0.78 0.70 0.76
1.04 1.33 1.20 1.33 1.32 1.12 1.54
0.49 0.56 0.84 0.68 1.01 0.84 0.71



Crash Frequency Deviation (Per Mile vs. Per VMT)
VMT (below) only using 26% of data!
VMT (below) only using 29% of data!
Pavement Condition
Good
Satisfactory
Fair
Poor
Very Poor
Serious
Failed
% Diff from Average
-21%
-6%
-4%
+19%
+30%
+40%
+38%
-10%
+2%
-1%
+9%
+4%
+1%
+7%
-26%
-13%
+2%
-7%
+48%
+77%
+120%
-30%
-17%
+7%
-17%
+33%
+52%
+53%
Crash Severity Deviation (Per Mile vs. Per VMT)
VMT (below) only using 26% of data!
VMT (below) only using 29% of data!
% Diff from Average
-19%
-3%
-3%
+13%
+28%
+35%
+47%
-16%
+8%
-3%
+8%
+7%
-9%
+25%
-15%
-18%
+3%
+1%
+43%
+37%
+66%
-32%
-22%
+16%
-6%
+41%
+17%
-1%
CL 4 LN COLLECTOR WITH 2 WY LEFT TURN LN (Per Mile)
CL 4 LN COLLECTOR WITH 2 WY LEFT TURN LN (Per VMT)
MJ SIX LANE URBAN MAJOR (Per Mile)
MJ SIX LANE URBAN MAJOR (Per VMT)
Street Type & Calculation Method



+
−
cartodbdarkmatter
Zoning Districts
Road Conditions (2016)
Soledad Freeway
San Clemente Canyon Freeway
San Diego Freeway
Jacob Dekema Freeway
Escondido Freeway

2016


+
−
cartodbdarkmatter
Zoning Districts
Road Conditions (2023)
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2023

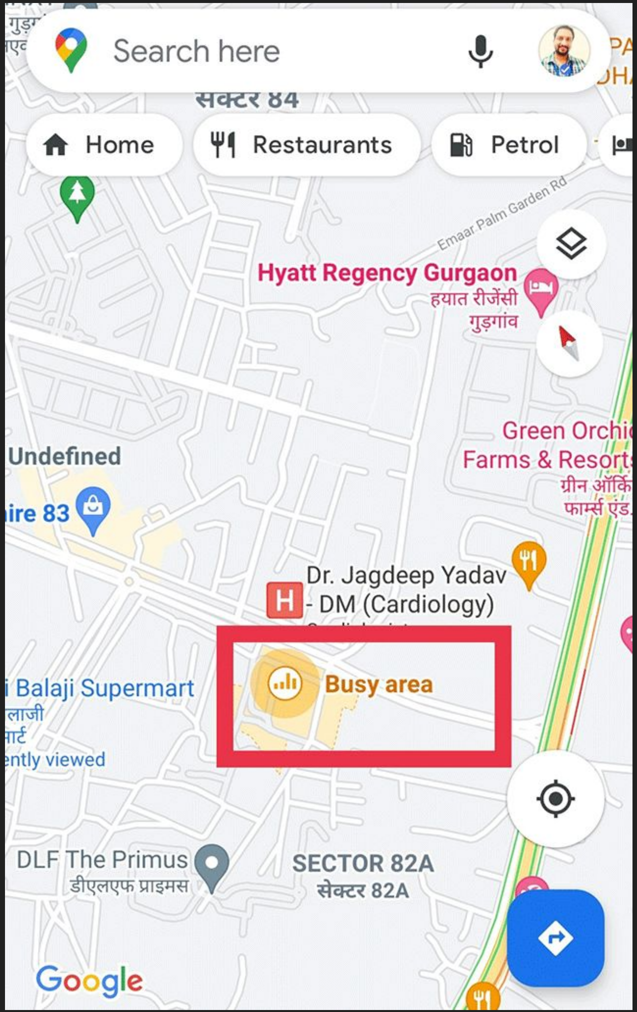
What are the Hotspots in San Diego?

How can we find Hotspots? (e.g. **Busy area** on Google Map)

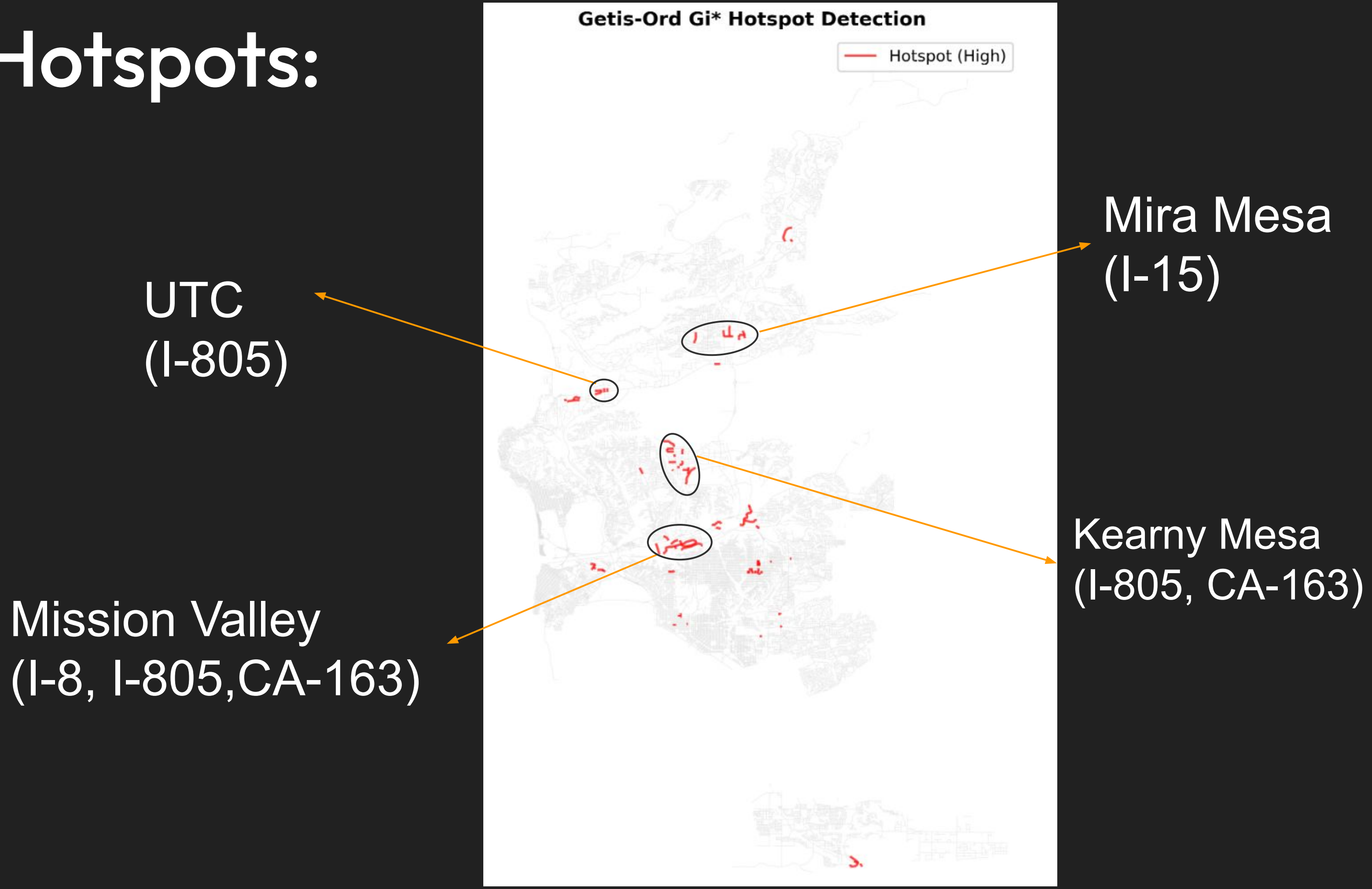
- Relatively **high** traffic volumes
- Relatively **more collisions**

Why we need Hotspots?

- Reference for incoming road projects
- Where do San Diegans usually go?



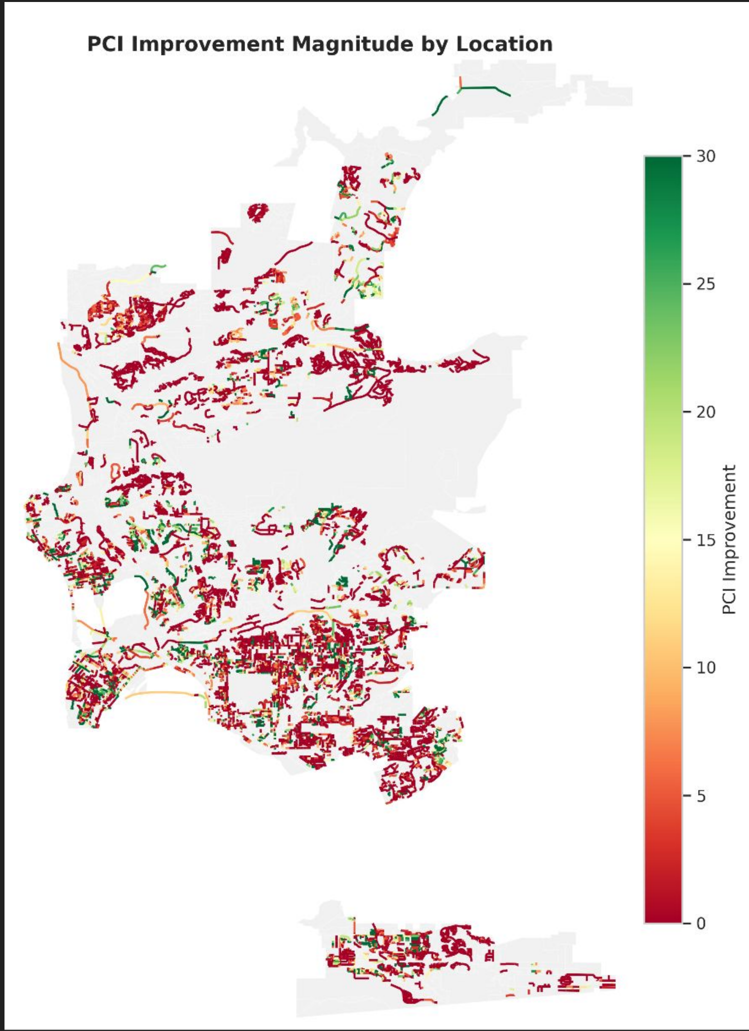
Traffic Hotspots:

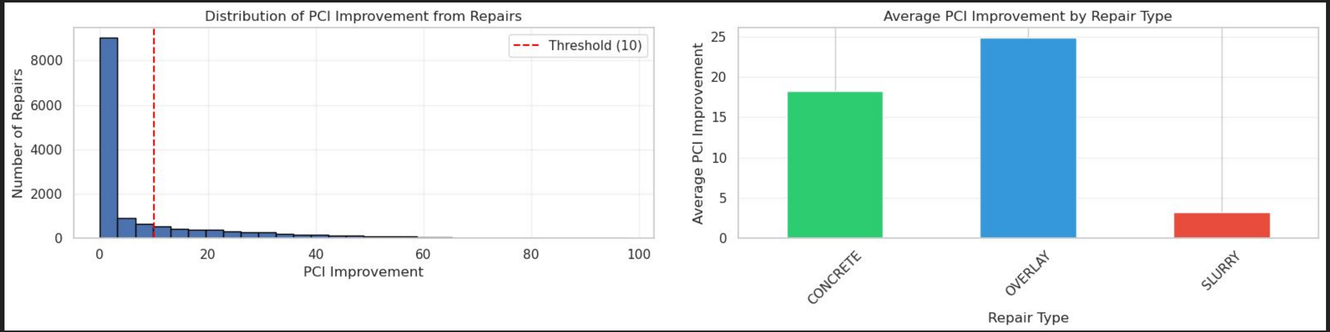


Are Road Repairments in SD Effective?

- We use **Pavement Condition Index (PCI)** to indicate pavement condition.

- Most Repairments improve PCI by a limited rate. Why?





Repair Type	Count	Avg Improvement	Median Improvement	Effectiveness Rate	Avg PCI Before	Avg PCI After
OVERLAY	2,900	**24.85**	23.04	**73.0%**	62.58	87.44
CONCRETE	104	**18.21**	13.97	**62.0%**	71.56	89.77
SLURRY	11,128	**3.24**	0.00	**12.0%**	84.26	87.49

OVERLAY: most effective → *For PCI < 65*

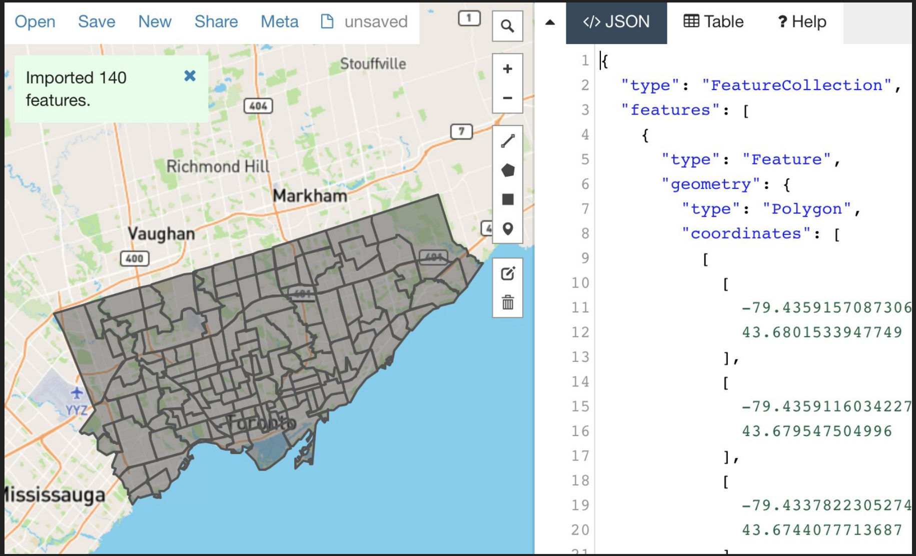
CONCRETE: also effective, but higher cost → *For critical infrastructures*

SLURRY: minimal improvement, but typically start from higher PCI, just preventive maintenance. → *For PCI > 80 and maintenance*

Reasonable and Effective!

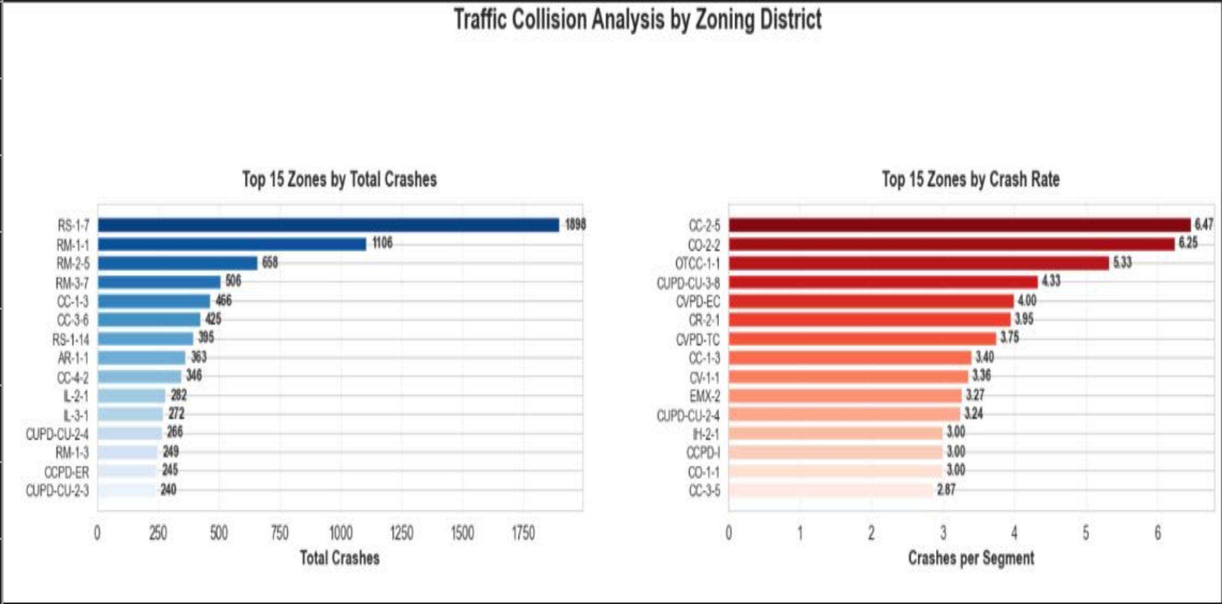
How do collision densities vary in different zones?

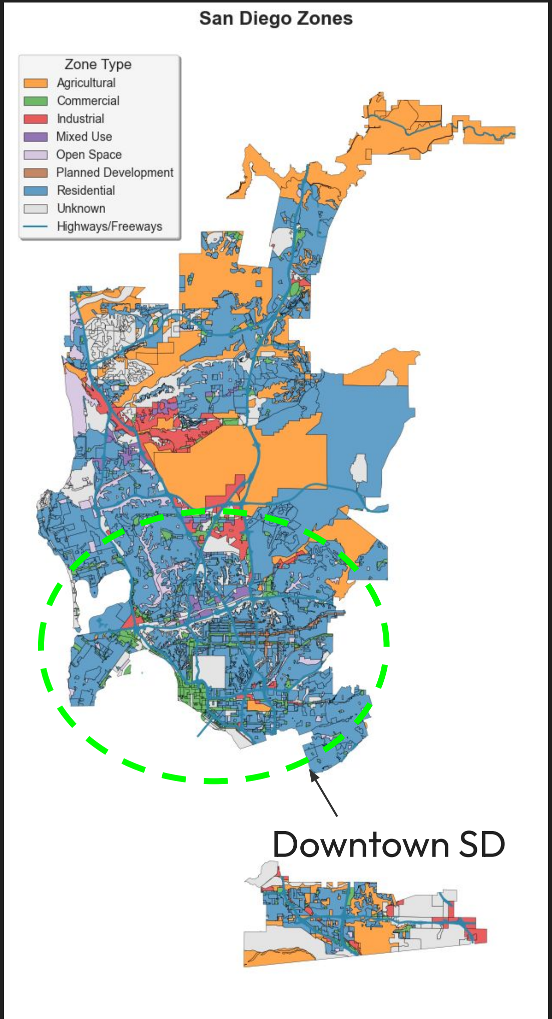
- **Zone:** A geographic polygon from SD's official zoning data (.geojson)
- Examples include residential (RS, RM), commercial (CC, CN, CCPD)
- Explore how different land uses correlate with traffic patterns
- Compute overlapping coordinates between collision sites and zone polygons

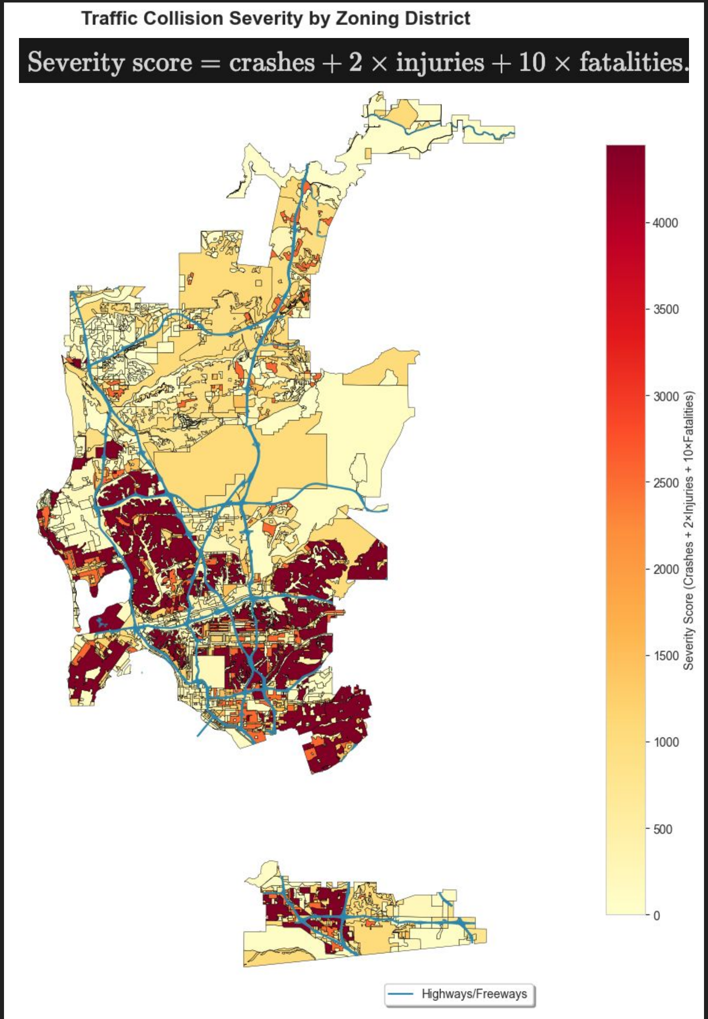


Processing a .geojson file

Most Problematic Zones		
Rank	Name	Type
1	RS-1-7	Residential Single-Family
2	RM-1-1	Residential Multi-Family
3	RM-2-5	Residential Multi-Family
4	RM-3-7	Residential Multi-Family
5	CC-1-3	Commercial
6	CC-3-6	Commercial
7	RS-1-14	Residential Single-Family
8	AR-1-1	Agricultural
9	CC-4-2	Commercial
10	IL-2-1	Industrial Limited







Residential and commercial zones most prone to accidents!

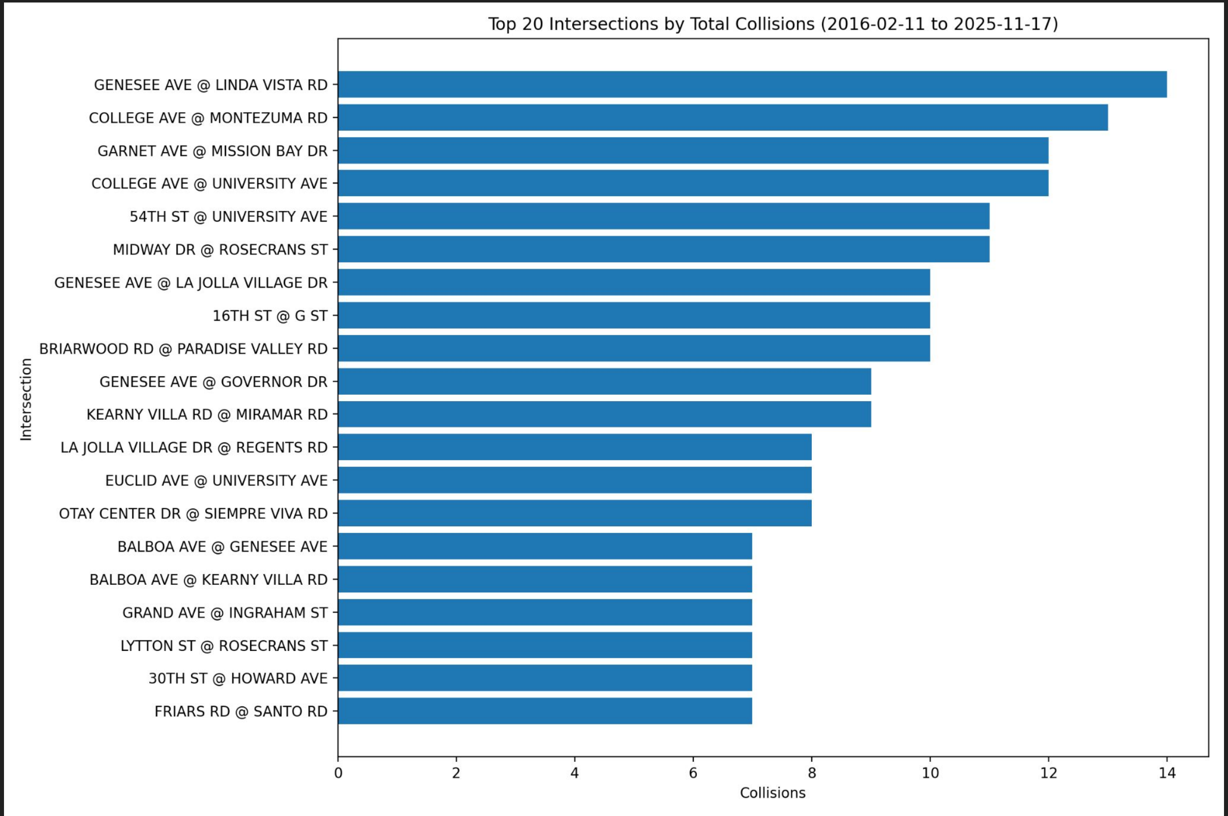
What are the most dangerous intersections in San Diego?

- How to define dangerous?
  - Raw number
  - By VMT (vehicle miles traveled)
- Why are these intersections dangerous?

Dangerous intersections by raw number of collisions

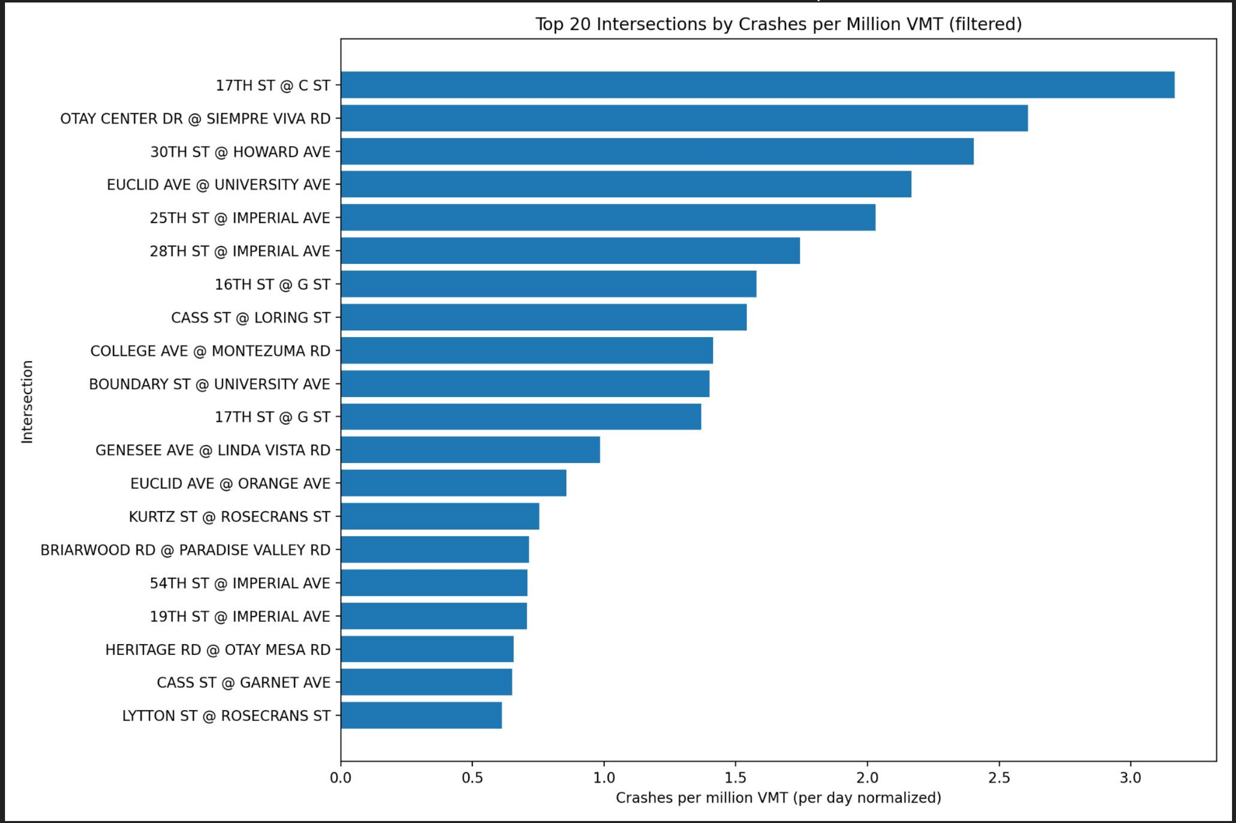
Why?

- **Near Colleges**
  - **Genesee Ave**
  - **University Ave**
  - **College Ave**
- **Major corridors in SD**
- **Near major Freeways**



Dangerous intersections by collisions per million VMT (Vehicle Miles Traveled)

- **8 intersections on both plots**
- **College Areas still apparent**
- **Imperial Ave 4 times**
- **Downtown streets more apparent**



Thank You for Listening!